

Farjana Kabir

Applied ML Engineer — Gen AI & LLM Specialist

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LinkedIn — GitHub

Summary

Applied ML Engineer (2+ years) specializing in **LLM infrastructure, RAG pipelines, embeddings, and agentic workflows**. Experienced in designing **production-ready Gen AI applications**, scalable deployment pipelines, and **vector-based knowledge systems**. Skilled in **Docker, CI/CD, FastAPI, and cloud deployment**, with proven success in improving system accuracy, optimizing retrieval, and enabling enterprise-wide AI adoption.

Core Skills

- **LLM Infrastructure:** Deployment (LLaMA, GPT, HuggingFace models), evaluation pipelines
- **Knowledge Systems:** RAG, embeddings, vector DBs (FAISS, ChromaDB, AstraDB), semantic search
- **Agentic Workflows:** LangChain, multimodal LLMs, function-calling APIs, autonomous assistants
- **MLOps & Deployment:** Docker, FastAPI, AWS, CI/CD, monitoring, scalable pipelines
- **Data Handling:** Tokenization, chunking strategies, preprocessing (Python, Pandas, NumPy)

Professional Experience

Senior Data Scientist

somikoron.ai

Sep 2023 – Feb 2025

- Deployed **RAG-based Bengali QA System** (embeddings + vector DB) → improved retrieval accuracy by **15%**.
- Built **multimodal prescription parser (LLM-powered)** → achieved **70% extraction accuracy**.
- Designed and deployed **YOLO-based object detection system** for government sector.
- Optimized **recommender pipelines** → **+10% performance**.
- Developed **custom Bengali tokenization & stemming workflow** → reduced vocabulary size by **3.34%**.

Data Scientist

somikoron.ai

Mar 2023 – Aug 2023

- Developed **domain-specific Bengali NLP applications** with embeddings + preprocessing.
- Engineered **data cleaning and ingestion pipelines** for large-scale knowledge extraction.
- Designed **web scraping + NLP parsing workflows** → extracted ~50 structured data points.

Key Projects

- **Agentic RAG QA System** – Modular pipeline with hallucination reduction (LangChain, Hugging Face, FastAPI).
- **Ecommerce Product Assistant** – Semantic search + embeddings with vector DB for retrieval optimization.
- **QuickSum** – Lightweight summarization system, benchmarked with ROUGE.
- **Bilingual QnA System** – English-Bengali assistant with **LoRA fine-tuned embeddings**.
- **GemAide Assistant** – Gemini-based multimodal app enabling **85% faster app creation**.

Education

B.Sc. in Computer Science & Engineering

KUET (2017 – 2022)

Concentration: Machine Learning

Relevant Coursework: NLP, Data Mining, ML, Image Processing, Biomedical Engineering

Publications

Diagnosis of Diabetic Retinopathy Using Deep Learning Techniques, *IEEE EICT 2021*.
DOI: 10.1109/EICT54103.2021.9733545

Certifications

- AI Engineer for Data Scientists – Associate (DataCamp)
- Associate Data Scientist (DataCamp)

Awards

Dean's Award – Khulna University of Engineering & Technology